

Response to Reviewers

GraphyloVar: Predicting the impact of non-coding variants
using a multi-species sequence model

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We thank the reviewers for their reading and constructive feedback. We addressed all comments and the revised manuscript marks all new or changed text in red.

Reviewer 1

R1.1 GCN interpretability and species-level analysis

The manuscript proposes the use of GCN to integrate multi-species sequence information in order to capture cross-species signals. However, the current presentation lacks a detailed analysis of this component. The author is encouraged to include additional investigations into the correlations among multi-species sequences and to enhance the interpretability of the sequence representations learned by the GCN module.

We added a new section (Section 3.6, GCN Interpretability and Species Importance) that provides a detailed additive perturbation analysis of the GCN component. For each of the 58 extant placental mammal species, we replaced all other alignment rows with a gap token and measured the decrease in nucleotide cross-entropy relative to an all-gap baseline, evaluated on 500,000 held-out variants from chromosomes 13 to 22. This quantifies the isolated information content of each species.

The results show that 53 of 58 extant species contribute positive signal. Human provides the largest contribution ($\Delta\text{CE} = 7.1$). Four diverged primates (Orangutan, Green monkey, Gorilla, Gibbon; $\Delta\text{CE} = 0.62$ to 1.50) form a secondary cluster. Chimpanzee ranks 11th ($\Delta\text{CE} = 0.062$) despite being the closest relative of humans because its sequence is nearly identical to the Human row, leaving no additional marginal signal (leave-one-out marginal $\Delta\text{CE} \approx -2.4 \times 10^{-6}$). A conditional perturbation analysis in which the Human row is also masked confirms this: Chimpanzee becomes the top non-human contributor in that setting. The SE-attention gate values learned by the model are consistent with these perturbation rankings, showing the model learns to weight species according to their phylogenetic informativeness rather than treating them equally.

Section 3.6 added; Figure 6 (species importance bar chart, clade-colored, log scale) and Supplementary Figures S1 (phylogenetic tree ablation) and S2 (model component ablation) added.

R1.2 Justification for 65 bp context window

In the Methods section, a fixed sequence fragment length of 65 bp is adopted for feature extraction, yet no sufficient justification for this parameter choice is provided. To strengthen the methodological rigor, the author should consider conducting systematic experiments to demonstrate the optimality or robustness of this setting.

We trained models with three context-window settings: flank $\in \{16, 32, 100\}$ bp (input sequences of 33, 65, and 201 bp). All models used the same training procedure on chromosomes 1 to 10 with chromosomes 11 to 12 as validation. Flank = 32 (65 bp) achieves the highest overall held-out AUROC (0.625) among the three settings, and also achieves the highest AUROC on all 13 MPRA benchmark datasets after fine-tuning, supporting this choice for downstream regulatory prediction tasks. These results are reported in Supplementary Table S1.

Section 3.5 and Supplementary Table S1 added.

R1.3 Quantitative evidence for complementarity

In the section “Complementarity of GraphyloVar Scores with Existing Methods,” the author claim that GraphyloVar identifies unique functional variant information that existing approaches fail to capture. However, this assertion is not supported by corresponding quantitative or qualitative experimental evidence. Additional analyses are needed to substantiate the claimed complementarity and to demonstrate the unique contributions of GraphyloVar.

We added quantitative evidence in Section 3.4. We computed a parameter-free z-score ensemble of GraphyloVar and CADD on the full ~ 149 M held-out variants. Both scores were independently z-normalized; because CADD assigns higher scores to more deleterious variants (i.e., rarer ones), its z-score was negated to align direction before averaging. The ensemble achieves AUROC = 0.6442, an improvement of +0.020 over GraphyloVar alone ($p < 10^{-15}$, DeLong test on 500,000 held-out variants). Bootstrap 95% confidence intervals ($B = 1,000$) for the ensemble and GraphyloVar alone do not overlap. The low Spearman correlation between GraphyloVar and CADD shown in Figure 4, together with this AUROC gain, confirms that GraphyloVar captures information that existing methods do not.

Section 3.4 expanded; Figure 5 added.

R1.4 Comparison with Evo2 and GPN-Star

Several existing approaches leverage multi-species sequence information to predict the functional effects of non-coding variants, including Evo2 and GPN-Star. These models have demonstrated strong performance in integrating evolutionary conservation with functional annotations. To more comprehensively assess the performance, novelty, and practical advantages of GraphyloVar, it is recommended that the author include a quantitative comparison with Evo2 and GPN-Star in the experimental evaluation section.

[1] Genome modeling and design across all domains of life with Evo 2.

[2] Predicting functional constraints across evolutionary timescales with phylogeny-informed genomic language models.

GPN-Star was added as a baseline in the common vs. rare variant AUROC benchmark (Figure 2A). Due to the computational cost of scoring $\sim 149\text{M}$ variants with GPN-Star, Spearman correlation with observed MAF was not computed for GPN-Star, so it does not appear in Table 1. GPN-Star was also not included in the MPRA evaluation because fine-tuning GPN-Star on MPRA data was outside the scope of this work.

Evo2 requires GPU compute capability 8.9 or higher (Ada or Hopper architecture). Our hardware consists of NVIDIA Quadro RTX 6000 GPUs (compute capability 7.5), which is below this requirement. We state this explicitly in Methods Section 2.4 and note Evo2 as an important future comparison in the Discussion.

Methods Section 2.4 and Discussion updated with GPN-Star paragraph and Evo2 hardware justification.

R1.5 Code and data availability

The author provides a code repository link in the manuscript; however, the repository does not include the complete implementation, nor does it contain the datasets used in the experiments. To ensure the reproducibility of the study, it is strongly recommended that the author release the full source code along with the corresponding datasets, and clearly specify the procedures for accessing them as well as their usage licenses.

A Data and Code Availability section was added at the end of the manuscript. It provides the GitHub URL for source code, trained model weights (flank = 32 checkpoint), and preprocessing scripts, released under the MIT License. It also provides the dbGaP accession number (phs000964) for the TOPMed whole-genome sequencing data, the UCSC URL for the 100-way vertebrate alignments, and pointers to the original publications for each MPRA dataset.

Data and Code Availability section added.

R1.6 Dataset descriptions and figure label consistency

On Page 6, Line 10, the manuscript mentions the use of “saturation mutagenesis of disease-associated regulatory elements [Kircher et al., 2019],” yet no direct comparison or explicit presentation of results based on this dataset is provided in the main text. In addition, Figure 3 and Figure 5 includes an annotation referring to the “GWAScatalog” for 3’ UTRs of genomic regulatory contexts, but the database is not described elsewhere in the manuscript. The author should supply the relevant dataset descriptions and ensure consistency between cited resources and the results presented.

All 13 MPRA datasets are now explicitly described in Methods Section 2.5, grouped by the three categories shown in Figure 3. Kircher et al. (2019) provided functional scores for all possible single-nucleotide variants (all three alternative alleles at each position) within disease-associated regulatory elements, forming part of the genome-wide analyses group along with Abell et al. (2022) and McAfee et al. (2023). For the 3’ UTR group, Griesemer et al. (2021) used annotations from the GWAS Catalog to identify causal 3’ UTR variants, and Schuster et al. (2023) characterized 3’ UTR variants associated with prostate cancer. The label used in Figures 3 and 5 refers to the GWAS Catalog as the annotation source for that dataset, and its meaning is now explained in the Methods text.

Methods Section 2.5 rewritten to describe all 13 MPRA datasets.

R1.7 MAF correlation reported only in text

The section “Correlation with Observed Minor Allele Frequencies” reports result only in textual form. To improve the clarity and credibility of the conclusions, it would be beneficial to include relevant supplementary figures or tables.

The Spearman correlation values are reported in Table 1 (Section 3.2). The table includes GraphyloVar and five baselines (GPN-MSA, Enformer, PhyloP, CADD, PhastCons). GPN-Star Spearman correlation was not computed as described in R1.4 above.

Table 1 added to Section 3.2.

R1.8 Figure 1 architecture inconsistency and white text

In Figure 1 (Page 4), the features after Mean Pooling are shown as an input to the “Scaling species with Attention” module. Since mean pooling typically produces a one-dimensional vector, this appears inconsistent with the subsequent GCN module, which seems to require high-dimensional feature inputs. The author should clarify the rationale behind this architectural design or revise the figure to accurately reflect the implemented model. Additionally, there is unexplained white text near the “Concatenate Left + Right” in the figure, the author should verify its validity and correct or remove it as appropriate.

Methods Section 2.2 and the Figure 1 caption were updated to clarify the pooling operation. After processing both the forward and reverse complement strands independently through two Transformer layers each, the output for each strand is mean-pooled over the 65 input positions, producing a 32-dimensional representation per strand. The forward and reverse complement strand representations are concatenated into a 64-dimensional feature vector per species (32 from the forward strand and 32 from the reverse complement strand), which is then passed to the SE block and GCN. This is consistent with the “Concatenate Left + Right” operation shown in Figure 1.

The white text near “Concatenate Left + Right” in the original figure is a rendering artifact from the figure generation tool and does not correspond to any model component.

Methods Section 2.2 and Figure 1 caption updated.

R1.9 Figure 2 caption label error

In Figure 2 (Page 7), the figure caption refers to “GraphyloVar Score (dark purple) and GraphyloVar Score (purple)” which appears redundant or incorrect. The author should carefully check the caption text, legend colors, and corresponding labels for accuracy and consistency.

The Figure 2 caption was rewritten to remove color references. The updated caption describes each panel directly without redundant or confusing color labels, since the figure legend already identifies each method.

Figure 2 caption rewritten.

R1.10 Naming consistency: GPN-MSA

There is inconsistency in the naming of the proposed method throughout the manuscript (e.g., “GPN-MSA” vs. “GPNMSA”). The author should standardize the nomenclature across the entire text and explicitly clarify whether the hyphen should be retained.

We use “GPN-MSA” (with hyphen) consistently throughout the manuscript. We verified this with `grep`; no instances of “GPNMSA” without a hyphen remain in any of the source files.

Global find-replace applied.

Reviewer 2

R2.1 Held-out data definition and potential leakage

Held-out data definition: The manuscript states that variants are “held out,” but it is unclear whether this refers only to variant-level exclusion or also to separation at the level of genomic regions. This should be stated explicitly to rule out potential leakage. If possible, an evaluation of region-based hold out would strengthen the work.

The holdout strategy is explicitly stated in Section 3.1. Chromosomes 1 to 10 were used for training, chromosomes 11 to 12 for validation, and chromosomes 13 to 22 exclusively for the held-out test evaluation. This chromosome-level partition ensures that no variant or genomic position from the test chromosomes appears anywhere in training or validation. The test set contains approximately 149 million SNVs from the TOPMed whole-genome sequencing cohort. This chromosome-level split provides separation at both the variant level and the genomic region level, because entire chromosomes are withheld.

For MPRA fine-tuning evaluation, leave-one-out cross-validation was used within each dataset group, as described in Section 3.3.

Section 3.1 first paragraph updated.

R2.2 Discussion of PHACT and PHACTboost

Prior work that incorporates phylogenetic information, such as PHACT (Kuru et al. 2022) and PHACTboost (Kuru et al., 2024), should be discussed to better link GraphyloVar within the phylogeny-aware variant prediction literature, even if the modeling approach differs or the work focuses on coding regions.

A comparison paragraph with PHACT (Kuru et al., 2022) and PHACTboost (Kuru et al., 2024) was added to the Discussion. PHACT and PHACTboost use protein-level features including amino acid conservation, protein structure, and functional domain annotations, and are designed specifically for missense variants in coding regions. GraphyloVar targets both coding and non-coding variants using only multi-species whole-genome alignments and does not use any protein-level features. The two approaches are therefore complementary: PHACT provides high accuracy for missense variant interpretation while GraphyloVar provides genome-wide coverage across all variant types. We note in the Discussion that combining them in an ensemble for coding variants would be an interesting direction for future work.

Discussion paragraph added.

R2.3 Comparison breadth relative to dbNSFP

The dbNSFP hosts 36 algorithms' prediction scores. Comparing to a larger set of tools should be possible, and if not, an explicit justification for their exclusion should be included.

A justification paragraph was added to Methods Section 2.4. We selected baselines representing methodologically distinct families: conservation metrics (PhyloP, PhastCons), integrative scoring (CADD), sequence-based deep learning (Enformer), and MSA-aware models (GPN-MSA, GPN-Star). Running all 36 dbNSFP tools was outside our computational scope; many share underlying conservation or annotation features, so the selected families collectively cover the major methodological landscape of variant effect prediction without redundancy.

Methods Section 2.4 updated with method selection justification.

R2.4 Loss function clarification

The allele-frequency head outputs a categorical distribution, yet binary cross-entropy is reported as the loss. This requires clarification.

We corrected this in Methods Section 2.3. The allele frequency head uses categorical cross-entropy loss because it predicts a probability distribution over five allele states (A, C, G, T, gap). The SNP probability head uses binary cross-entropy loss because polymorphism status is a binary label. The original text incorrectly stated binary cross-entropy for both heads.

Methods Section 2.3 corrected.

We thank the reviewers again for their careful reading and comments. We believe the revised manuscript fully addresses all the points raised.